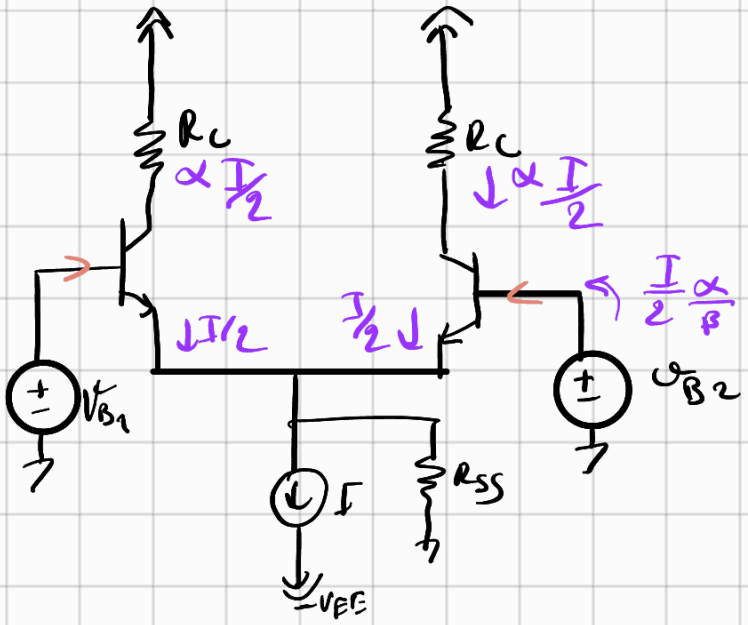
BJT Differential Amplifier

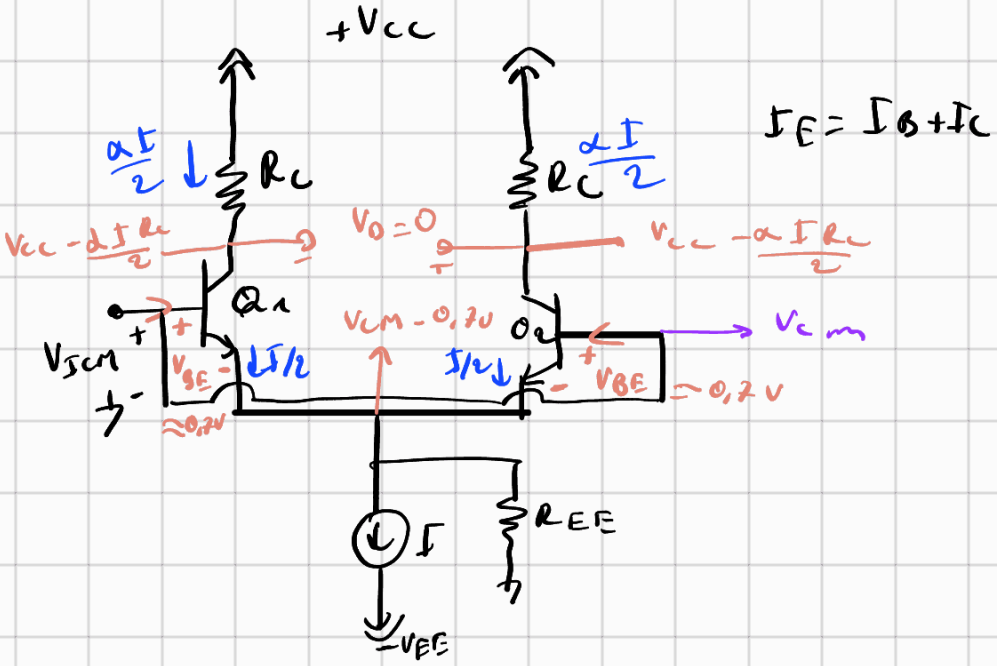


$I_B \neq 0$

$g_m = \frac{I_C}{V_T}$ → 25,8 mV at room temp

$r_\pi = (\beta+1) r_e$

$r_e \simeq 1/g_m = \frac{V_T}{I_E}$

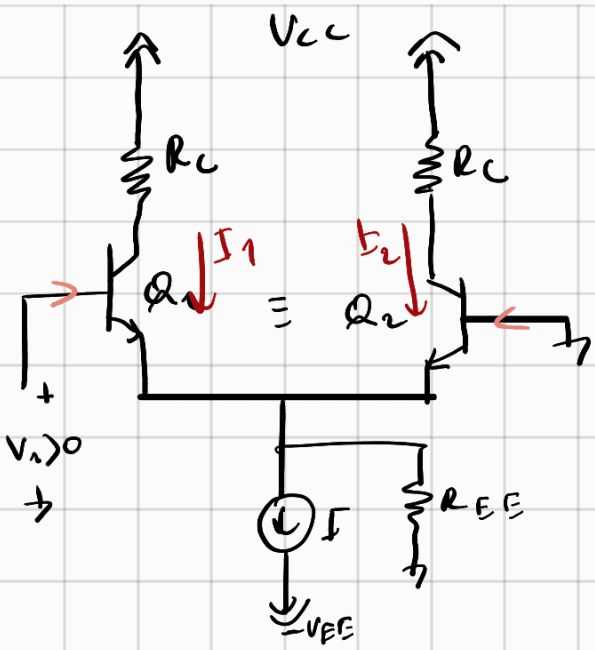


$I_E = I_B + I_C$

$Q_1 \equiv Q_2$

$(EBJ_{Area_1} = EBJ_{Area_2})$

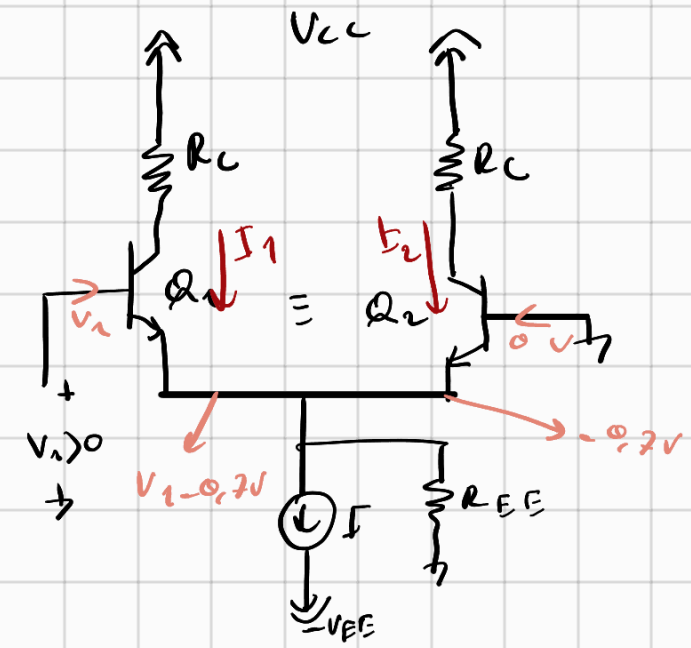
$I_C = I_S e^{V_{BE}/V_T}$



$I_1 = ?$

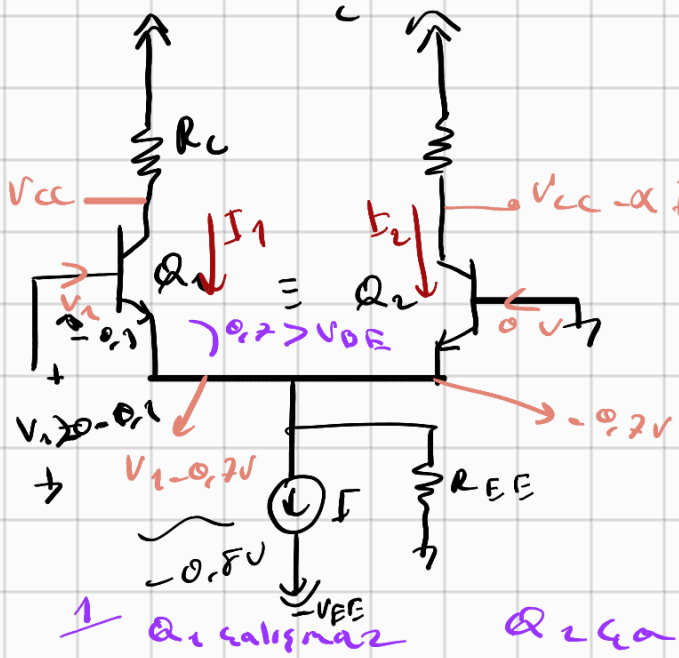
$I_2 = ?$

İki transistörün çalıştığını varsayalım



$V_1 - 0,7 \neq 0,2$ (Imkansız!)

(İkinciden kaynaklı)



$V_i = -0,1$ V olursa

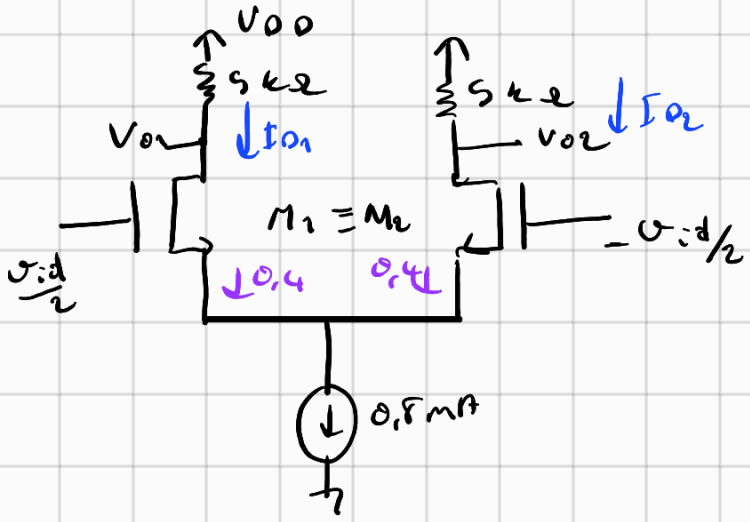
Q_1 çalışmaz

Q_2 çalışır

1 Q_1 çalışmaz Q_2 çalışır ✓

(1)

A MOS differential pair is operated at a bias current of 0,8mA. Find V_{ov}, $g_m r_o$, and A_{dif} using transistors with a $\frac{W}{L}$ ratio of 100, $\mu_n C_{ox} = 0,2 mA/V^2$, $V_A = 20V$ and $R_D = 5k\Omega$



$I_{D1} + I_{D2} = 0,8 mA$

since $M_1 = M_2$

$2 I_{D1} = 2 I_{D2} = 0,8 mA$

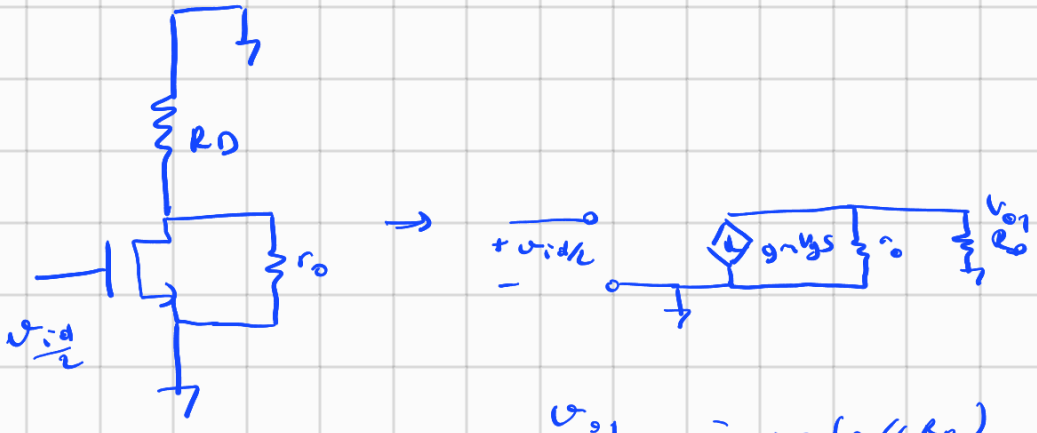
$I_{D1} = I_{D2} = I/2 = 0,4 mA$

$$0,4mA = \frac{1}{2} \cdot \frac{0,1}{L} \cdot \frac{10}{L} (V_{OV})^2$$

$V_{OV} = \begin{cases} +0,2 \\ -0,2 \text{ (kabul edilmez)} \end{cases}$

$$g_m = \frac{2 I_D}{V_{OV}} = \frac{2 \cdot 0,4}{0,2} = 4 mS$$

$$r_o = \frac{V_A}{I_D} = \frac{20}{0,4} = 50 k\Omega$$



$$\frac{v_{o1}}{v_{id/2}} = -g_m (r_o \parallel R_D)$$

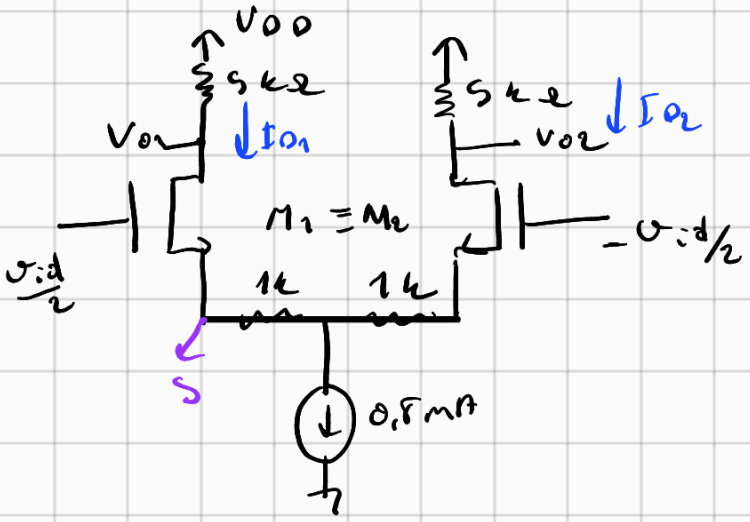
$$= -4 mS \, (50k \parallel 5k) \quad (4,54)$$

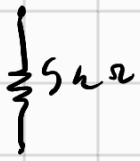
$$\frac{V_{o2}}{v_{id}} = 9,08 \left[\frac{V}{V}\right] \Leftarrow \frac{v_{o1}}{v_{id}} = 9,08 \left[\frac{V}{V}\right]$$

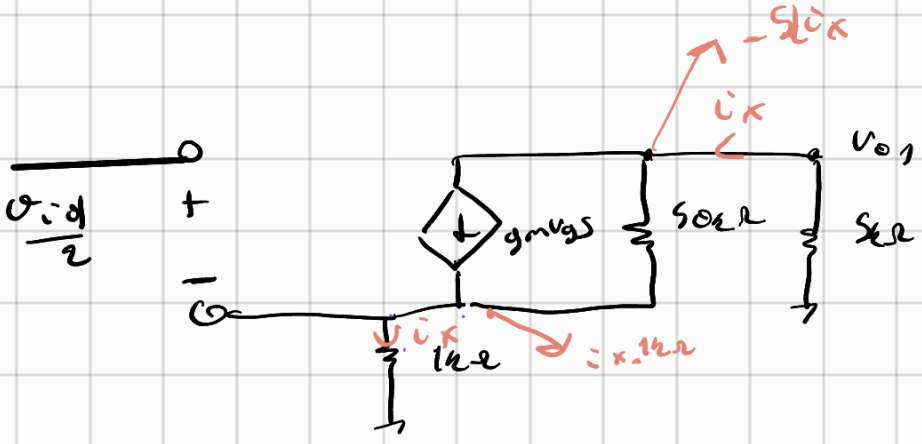
$$A_{diff} = \frac{v_{diff}}{v_{id}} = \frac{v_{o2} - v_{o1}}{v_{id}} = 18,16 \left[\frac{V}{V}\right]$$

$$= 20 \log(18,16) \text{ dB}$$

$$A_{diff} = 25,15 \text{ dB}$$







$$\frac{v_{o1}}{\frac{v_{id}}{2}} = \frac{v_{o1}}{v_{gs}} = ?!$$

$$v_{id} ?! \; v_{gs}$$

$$v_{o1} = -5k\,i_x$$

$$g_m v_{gs} + \left(\frac{-5i_x - i_x}{50}\right) = i_x$$

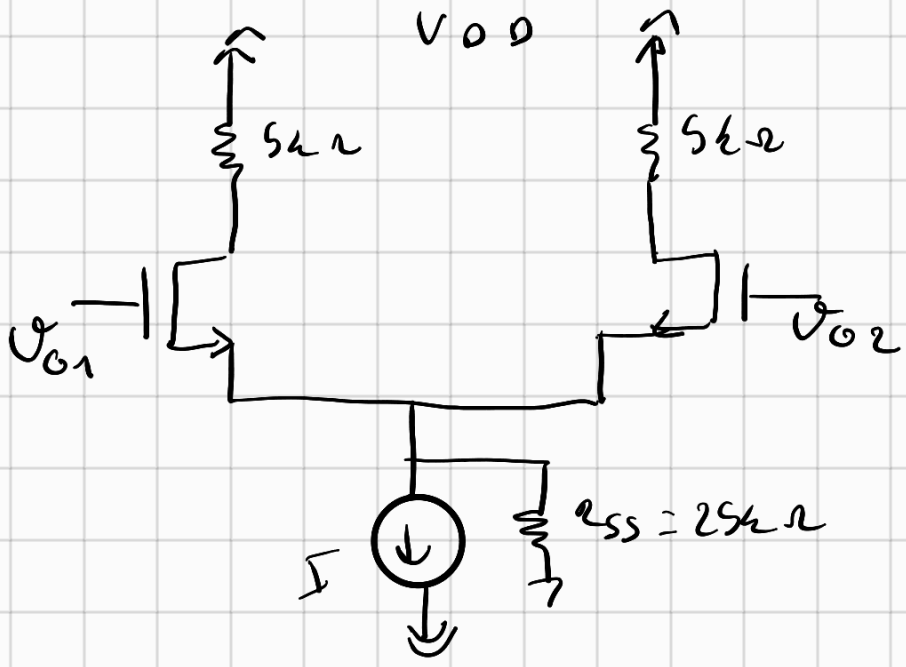
$$g_m v_{gs} = \frac{56 i_x}{50}$$

$$V_{gs} = 1/g_m \cdot \frac{56\,k\Omega}{50}$$

$$\frac{v_{o1}}{v_{gs}} = \frac{-56\,k\Omega}{\frac{56\,k\Omega}{200}} = -17,86\ [V/V]$$

$V_A = \infty$

$|A_{single}| = \frac{v_{o1}}{v_{id}/2}$

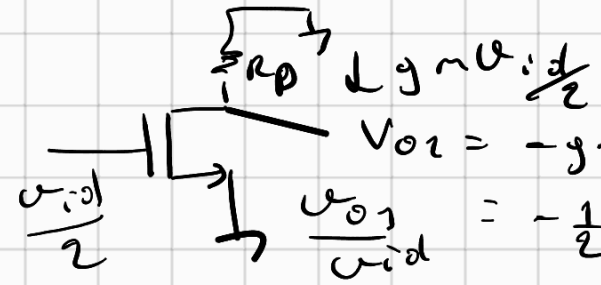


a) Find the differential gain, the common mode gain, and the common-mode rejection ratio if the output is taken single endedly. Assume devices are perfected matched

b) Repeat a for differential outputs

c) Repeat b when the resistances ha a mismatch of 1%

Diff mode half circuit model



$v_{o1} = -g_m \frac{v_{id}}{2} R_D$

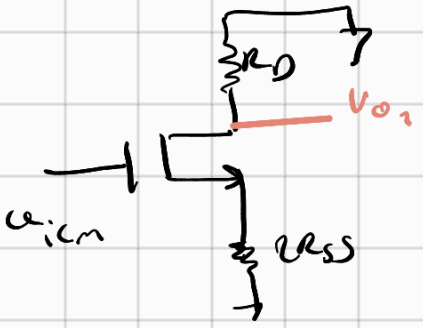
$\frac{v_{o1}}{v_{id}} = -\frac{1}{2} \cdot 4mS \cdot 5k\Omega$

$= -10 V/V$

$|A_{single}| = 10[V/V] = 20\,dB$

$|A_{diff}| = 40\,dB$

Common mode half circuit model



$\frac{v_{o1}}{v_{icm}} \simeq -\frac{R_D}{2R_{SS}} = \frac{-5}{50k}$

$= -0.1[V/V]$

$|A_{cm}| = +0.1[V/V] = -20\,dB$ //

$|A_{cm\,diff}| = 0$

$CMRR = \frac{A_{single}[V/V]}{A_{cm}[V/V]} = A_{single}[dB] - A_{cm}[dB]$

$= 40\,dB$

$CMRR = \infty$

$|A_{cm\,diff}| = |A_{cm}|_{single} \times \frac{1}{100}$

$= 1 \times 10^{-3}[V/V]$

$|A_{cm}|_{dB} = -60\,dB$

$CMRR_{total\,mismatch} = 40 - (-60)$

$= 100\,dB$